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Personal-Data Discovery and Privacy Evidence Inventory

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Problem Description

- Personal data builds up over time in spreadsheets and database tables, often with no documentation of why it is kept.
- Data protection frameworks expect organizations to know what personal data they hold, why they hold it, and how long they keep it.
- Today this is usually checked by hand, file by file, and depends on whoever knows the data best.
- Fields with no recorded purpose or retention period are easy to miss.
- Detection quality is rarely measured, so nobody knows how much a manual check misses.

Our Task: Scan synthetic CSV files and one SQL database, find candidate personal-data fields across five data classes, link them to declared purpose and retention metadata, flag what is missing, and produce an evidence inventory for human review.

Boundary: The tool collects technical evidence only. It does not decide lawful basis, judge legal compliance, or infer data flows beyond what it scans.

Background

- Personally identifiable information (PII) is data that can identify a person, alone or combined with other data.
- Two ideas recur in data protection frameworks: *purpose limitation* (data is kept for a stated purpose) and *storage limitation* (data is kept only as long as needed).
- To check either, an organization first needs an inventory of which fields hold personal data, with a purpose and retention period recorded for each.
- Finding those fields is hard. Column names can mislead, formats vary, and names and sensitive attributes are difficult to recognize from values alone.
- Because of this, our scanner reports candidate fields with the evidence behind each flag. It does not claim certainty, and a person makes the final call.
- An evidence inventory is a record for reviewers. It is not a legal verdict.

Literature & RFC Review

- MeitY data protection framework [1]: It gives the Indian regulatory context for knowing what personal data is stored and why.
- Digital Personal Data Protection Act, 2023 [2]: India's data protection law. It is the context for purpose and storage limitation, and we do not interpret it legally.
- GDPR Article 30 [3]: requires records of processing activities, including purposes and, where possible, retention periods. It is a useful model for our inventory fields.
- Microsoft Presidio [4]: an open-source PII detection tool that combines patterns, checksums and context. It is a good reference for detection, but it focuses on finding PII rather than linking it to purpose and retention records.
- NIST Privacy Framework [5]: includes inventory and mapping of data processing, which supports the idea of a field-level inventory.

Gap we address: Existing work either describes what should be documented or detects PII. Few lightweight tools do both and report how well they detect.

Proposed Solution

Sakshyan works in 5 steps

1. **Ingest:** read CSV files and one SQL database in read-only mode. Collect table names, column names, data types and limited sample values.
2. **Detect:** apply configurable rules (column-name patterns, regular expressions, format and checksum checks) and record which rule triggered each detection.
3. **Map Metedata:** match each field to a declared registry of business purpose and retention period. Flag anything missing
4. **Build the Evidence Inventory:** one record per candidate field, covering source, class, triggering rule, confidence, purpose, retention, gaps and review status.
5. **Evaluate and Report:** compare results with a manually labelled test set and report precision and recall for each class, plus an evidence gap report

Proposed Data Classes:

- Identity
- Contact
- Government and unique identifiers
- Financial
- Health and sensitive attributes

Configurable Evidence Checklist: a setting file lists the evidence each field needs (purpose, retention, owner)
Out of Scope: lawful-basis decisions and legal compliance judgements

Stack: Python, pandas, SQLite, FastAPI, GitHub

Optimization Proposed

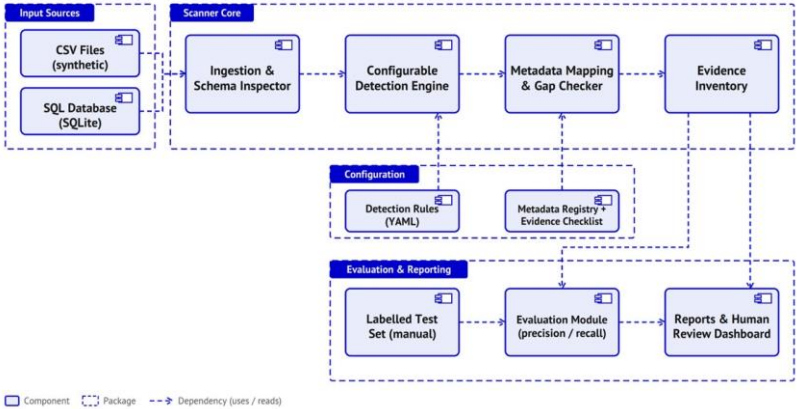
Manual Approach	Sakshyan
Inspect files by hand	Repeatable automated scan
Depends on individual knowledge	Configurable detection rules
Purpose and retention checked separately	Matched to one metadata registry
Missing documentation goes unnoticed	Missing metadata flagged
Detection quality not measured	Per-class precision and recall
Findings scattered	One exportable evidence inventory

Planned Enhancements:

- Checksums checks to cut false positives
- Confidence level and triggering rule shown for every flag
- Vague or missing metadata entries flagged
- Masked sample values in reports

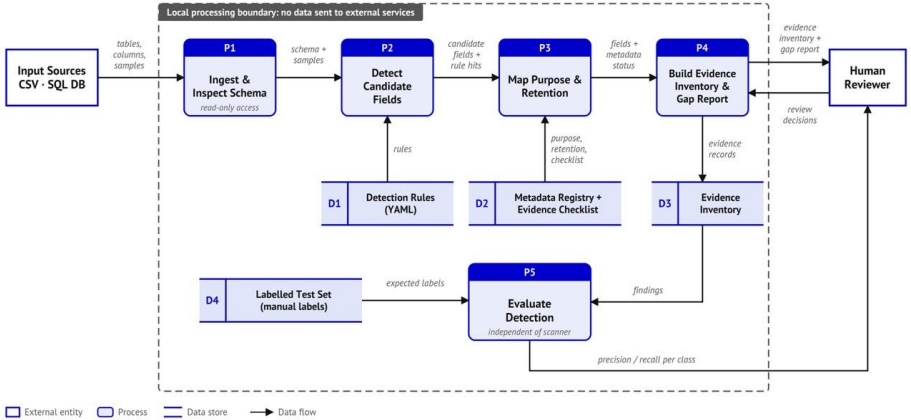
Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.



Deployment Timeline

Methodology: Iterative development in a shared GitHub repository, with regular mentor reviews

Timeline:

- **Phase 1**: finalise the data classes, rules format and metadata registry format with the mentor, Create the synthetic datasets and the labelled test set.
- **Phase 2**: build ingestion and the detection engine
- **Phase 3**: build metadata mapping, the evidence checklist and the gap report
- **Phase 4**: build the evaluation module and the dashboard.
- **Phase 5**: test, write the technical report, and prepare the demo with the pseudocode snippet

Bibliography

[1] Ministry of Electronics and Information Technology (MeitY), Government of India. Data Protection Framework. https://www.meity.gov.in/data-protection-framework

[2] Government of India. The Digital Personal Data Protection Act, 2023.

[3] European Union. General Data Protection Regulation (EU) 2016/679, Article 30.

[4] Microsoft. Presidio: Data Protection and De-identification SDK. [add URL]

[5] National Institute of Standards and Technology. NIST Privacy Framework, Version 1.0 (2020).

Thank You For Your Attention

Team Name: cipherHelix